1 Monotonic Sequences

Definition 1.1 (Increasing, decreasing, monotonic sequence)**.**

(a) A sequence $\{a_n\}$ is *increasing* if $a_n < a_{n+1}$ for all $n \geq 1$.

(b) A sequence $\{a_n\}$ is *decreasing* if $a_n > a_{n+1}$ for all $n \geq 1$.

(c) A sequence $\{a_n\}$ is *monotonic* if it is either increasing or decreasing.

1. Determine whether the sequence $\{\frac{3}{n+5}\}$ is increasing or decreasing.

2. Determine whether the sequence $a_n = \frac{n}{n^2+1}$ is increasing or decreasing. (*Hint*: If we define $f(x) = \frac{x}{x^2+1}$ for $x \in (1,\infty)$, then we have $a_n = f(n)$ for all $n \in \mathbb{N}$. Since f is differentiable, we can use f' to determine whether f is increasing or decreasing.)

2 Bounded Functions

Definition 2.1 (Bounded sequence)**.**

(a) A sequence $\{a_n\}$ is *bounded above* there is a number M such that $a_n \leq M$ for all $n \geq 1$.

(b) A sequence $\{a_n\}$ is *bounded below* there is a number m such that $m \leq a_n$ for all $n \geq 1$.

(c) A sequence $\{a_n\}$ is *bounded* if it is bounded above and bounded below.

3. Show that a sequence $\{a_n\}$ is bounded if and only if there exists a number M such that $|a_n| \leq M$ for all $n \geq 1$.

Example 2.2. Consider the sequence $a_n = n$. The sequence is bounded below, since $a_n = n \geq 1$. Suppose the sequence is bounded above. Then the set $\{a_n\}$ has a least upper bound, M. Since $M-1 < M$, $M-1$ is not an upper bound for the set $\{a_n\}$, so there exists N such that $M-1 < a_N = N$. But then $M < N+1 = a_{N+1}$, contradicting the assumption that M is an upper bound for the set $\{a_{N+1}\}$. Hence, the sequence is not bounded above.

4. Show that the sequence $a_n = \frac{n}{n+1}$ is bounded.

Theorem 2.3 (Every convergent sequence is bounded)**.** *If $\{a_n\}$ is convergent, then it is bounded.*

5. Prove Theorem 2.3. (*Hint*: Use the definition of convergence.)

6. Is the converse of Theorem (2.3) true? That is, is every bounded sequence convergent? (*Hint*: Consider the sequence $\{(-1)^n\}$.)

7. Is every monotonic sequence convergent? (*Hint*: Consider the sequence $a_n = n$.)

3 Bounded Monotonic Sequence Theorem

We have just seen that not every bounded sequence converges, and not every monotonic sequence converges. However, the next theorem shows that if a sequence is both bounded and monotonic, then it must converge. The figure below shows intuitively why this must be true. If $\{a_n\}$ is increasing and $a_n \leq M$ for all n, then the terms are forced to crowd together and approach some number $L \leq M$.

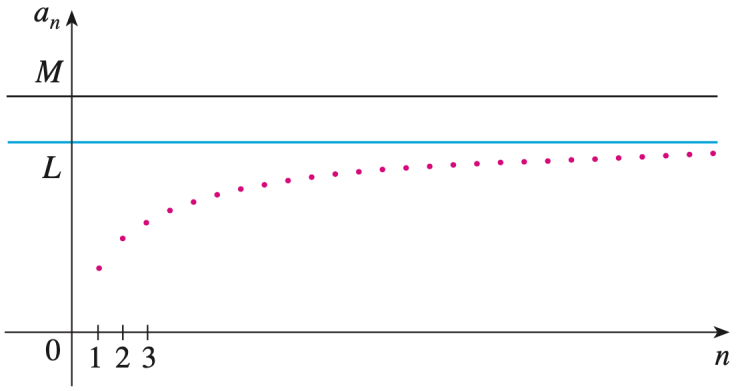


Figure 1: A bounded monotonic sequence is convergent.

Theorem 3.1 (Bounded Monotonic Sequence Theorem)**.** *Every bounded monotonic sequence converges. If* $\{a_n\}$ *is increasing, then* $\lim_{n\to\infty} a_n = \sup\{a_n : n \geq 1\}$. *If* $\{a_n\}$ *is decreasing, then* $\lim_{n\to\infty} a_n = \inf\{a_n : n \geq 1\}$.

Proof. Suppose $\{a_n\}$ is a bounded increasing sequence. Let L be the least upper bound of the set $\{a_n : n \geq 1\}$. Given $\varepsilon > 0$, $L - \varepsilon$ is not an upper bound for $\{a_n : n \geq 1\}$, so there exists $N \in \mathbb{N}$ such that $a_N > L - \varepsilon$. Since the sequence is increasing, $a_n \geq a_N$ for every $n \geq N$, and therefore $n \geq N$ implies

$$a_n > L - \varepsilon$$

and therefore

$$L - a_n < \varepsilon.$$

Since $\{a_n\}$ is increasing, $a_n \leq L$ for all $n \geq 1$ [1] and therefore $L - a_n = |L - a_n|$. Putting these together, $|L - a_n| < \varepsilon$ for $n \geq N$, which shows that $\lim_{n\to\infty} a_n = L$. □

[1] Proof: Suppose there exists N such that $L < a_N$. Then corresponding to $\varepsilon = a_N - L > 0$, there exists $M \in \mathbb{N}$ such that $n \geq M$ implies $|a_n - L| < \varepsilon$. Then for all $n > \max\{N, M\}$, we have $a_n - L \leq |a_n - L| < \varepsilon = a_N - L$ and therefore $a_n < a_N$, which contradicts the assumption that $\{a_n\}$ is increasing since $n > N$. Hence, $a_n \leq L$ for all $n \geq 1$. □

8. (Optional) Prove the bounded monotonic sequences theorem for $\{a_n\}$ a bounded decreasing sequence.

9. Use the Bounded Monotonic Sequence Theorem to determine whether the following sequences converge.

   (a) $a_n = 1 - \dfrac{1}{n}$

   (b) $a_n = 1 + \dfrac{1}{n}$

   (c) $a_n = \dfrac{n^{3/2}}{n^{3/2}+1}$

   (d) $a_n = \dfrac{(-1)^n n^{3/2}}{n^{3/2}+1}$